Examen de Tarification Automobile — BASEAUTO

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1 Avant de commencer

```
# setwd("C:/Users/VotreNom/Desktop/Examen_Tarification_2025_nom")
getwd()
```

```
[1] "C:/Users/SSQ/Pictures"
```

---

2 Partie A — Analyse exploratoire

2.1 Importation et visualisation des données

```
df <- read.csv("BASEAUTO.csv", sep = ";", dec = ".", stringsAsFactors = FALSE)
cat("Dimensions :", nrow(df), "lignes x", ncol(df), "colonnes\n")
```

```
Dimensions : 10496 lignes x 17 colonnes
```

```
head(df, 6) |>
  kbl_out(caption = "Aperçu des 6 premières observations")
```

Table 1: Aperçu des 6 premières observations

PERMIS	ACV	SEX	STATUT	CSP	USAGE	AGECOND	K8000	RM	CAR	CLA	ALI	ENE	VIT	SEGM	GARAGE	CHARGE
245	10	F	C	50	2	40	N	64	BER	A	CAR	ES	130-140	A	3	0
348	10	F	A	50	1	63	N	50	BER	A	CAR	ES	001-130	0	3	0
16	10	F	C	26	2	20	N	95	BER	A	CAR	ES	001-130	0	3	0
291	10	F	A	50	1	56	N	50	BER	A	CAR	ES	001-130	0	3	0
123	10	F	A	50	1	29	N	95	BER	A	CAR	ES	140-150	B	3	0
295	10	F	A	37	1	43	O	50	BER	A	CAR	ES	001-130	A	3	0

```
str(df)
```

```
'data.frame':   10496 obs. of  17 variables:
 $ PERMIS : int  245 348 16 291 123 295 24 181 157 338 ...
 $ ACV    : int  10 10 10 10 10 10 10 9 10 10 ...
 $ SEX    : chr  "F" "F" "F" "F" ...
 $ STATUT : chr  "C" "A" "C" "A" ...
 $ CSP    : int  50 50 26 50 50 37 50 50 55 1 ...
 $ USAGE  : int  2 1 2 1 1 1 2 3 1 2 ...
 $ AGECOND: int  40 63 20 56 29 43 21 35 31 48 ...
 $ K8000  : chr  "N" "N" "N" "N" ...
 $ RM     : int  64 50 95 50 95 50 90 50 72 50 ...
```

```
 $ CAR    : chr  "BER" "BER" "BER" "BER" ...
 $ CLA    : chr  "A" "A" "A" "A" ...
 $ ALI    : chr  "CAR" "CAR" "CAR" "CAR" ...
 $ ENE    : chr  "ES" "ES" "ES" "ES" ...
 $ VIT    : chr  "130-140" "001-130" "001-130" "001-130" ...
 $ SEGM   : chr  "A" "0" "0" "0" ...
 $ GARAGE : int  3 3 3 3 3 3 3 3 3 2 ...
 $ CHARGE : int  0 0 0 0 0 0 0 0 0 179 ...
```

Interprétation : La base BASEAUTO contient **10496 polices** et **17 variables** : caractéristiques du conducteur (âge, sexe, statut, CSP, ancienneté du permis), du véhicule (type, classe SRA, énergie, vitesse, segment) et la variable cible **CHARGE** (montant annuel de remboursement en euros). Le taux de sinistralité est de **13.3%**.

2.2 Conversion des types de données

```
vars_quali <- c("SEX","STATUT","K8000","CAR","CLA","ALI","ENE","VIT","SEGM","USAGE","GARAGE")
for (v in vars_quali) df[[v]] <- as.factor(df[[v]])

# Tableau récapitulatif
data.frame(
  Variable  = vars_quali,
  Modalites = sapply(vars_quali, function(v) nlevels(df[[v]]))
) |> kbl_out(caption = "Nombre de modalités par variable qualitative")
```

Table 2: Nombre de modalités par variable qualitative

	Variable	Modalites
SEX	SEX	2
STATUT	STATUT	2
K8000	K8000	2
CAR	CAR	9
CLA	CLA	26
ALI	ALI	5
ENE	ENE	3
VIT	VIT	10
SEGM	SEGM	6
USAGE	USAGE	4
GARAGE	GARAGE	3

2.3 Statistiques descriptives univariées

2.3.1 Variables quantitatives

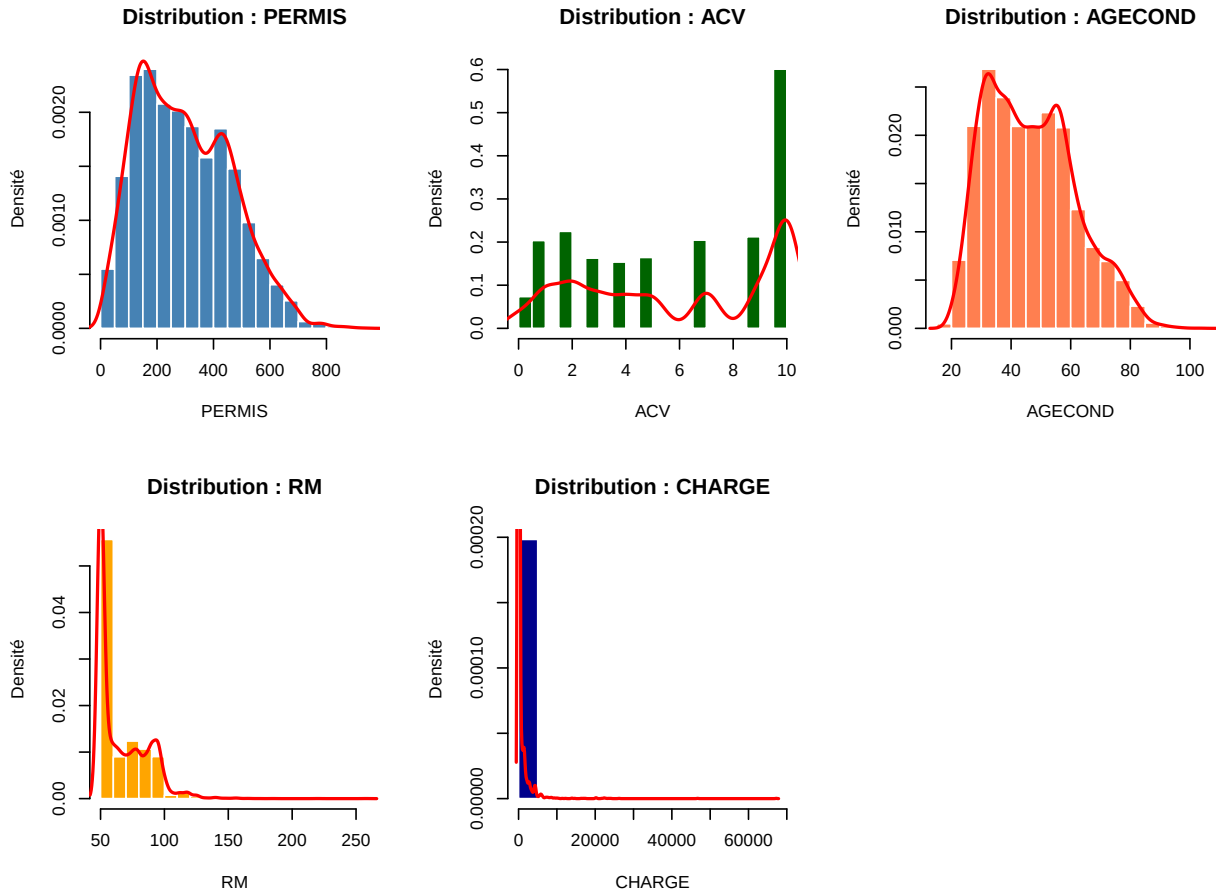
```
vars_quanti <- c("PERMIS","ACV","AGECOND","RM","CHARGE")

do.call(rbind, lapply(vars_quanti, function(v) {
  x <- df[[v]]
  data.frame(
    Variable   = v,
    Moyenne    = round(mean(x, na.rm=TRUE), 2),
    Ecart_type = round(sd(x, na.rm=TRUE), 2),
    Min        = round(min(x, na.rm=TRUE), 2),
    Q25        = round(quantile(x, 0.25, na.rm=TRUE), 2),
    Mediane    = round(median(x, na.rm=TRUE), 2),
    Q75        = round(quantile(x, 0.75, na.rm=TRUE), 2),
    Q99        = round(quantile(x, 0.99, na.rm=TRUE), 2),
    Max        = round(max(x, na.rm=TRUE), 2)
  )
})) |> kbl_out(caption = "Statistiques descriptives - variables quantitatives")
```

Table 3: Statistiques descriptives — variables quantitatives

	Variable	Moyenne	Ecart_type	Min	Q25	Mediane	Q75	Q99	Max
25%	PERMIS	298.89	162.42	1	163.75	281	422	689.0	940
25%1	ACV	5.96	3.56	0	2.00	7	10	10.0	10
25%2	AGECOND	46.75	14.76	19	34.00	45	57	82.0	103
25%3	RM	65.82	19.36	50	50.00	57	80	118.0	258
25%4	CHARGE	244.02	1406.10	0	0.00	0	0	4395.4	67261

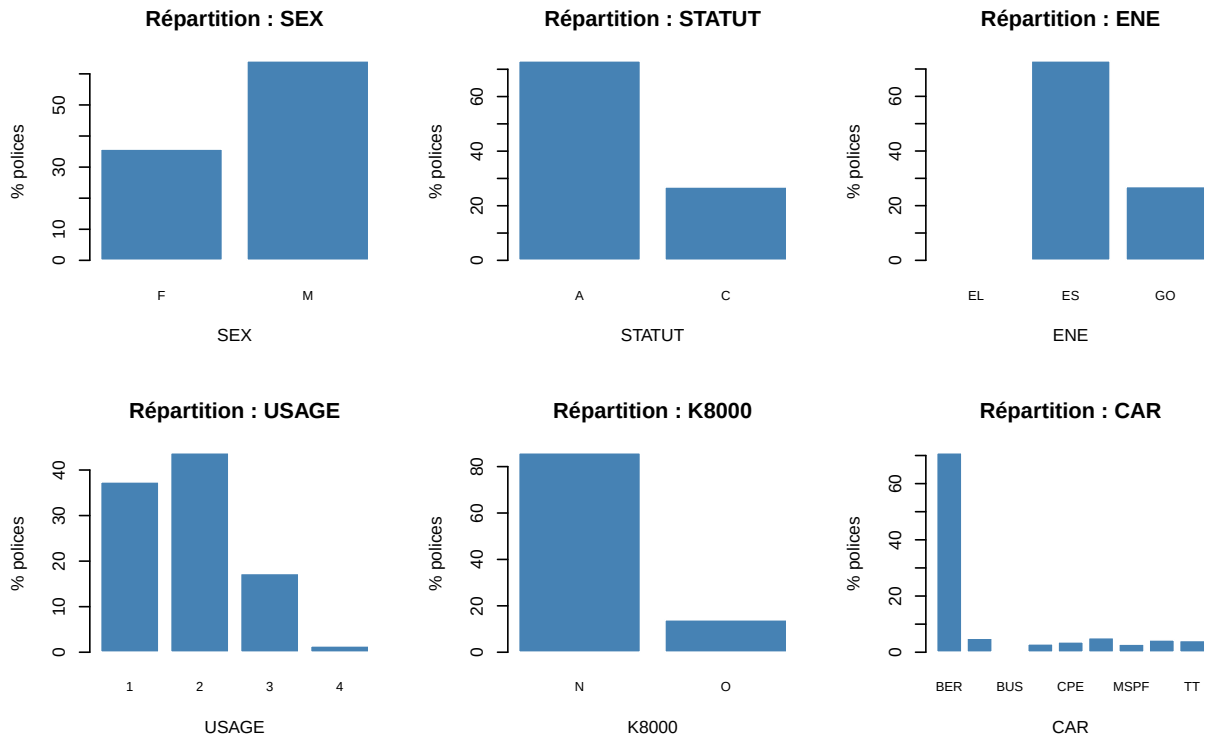
```
par(mfrow = c(2, 3))
cols <- c("steelblue","darkgreen","coral","orange","darkblue")
for (i in seq_along(vars_quanti)) {
  v <- vars_quanti[i]
  hist(df[[v]], probability = TRUE, col = cols[i], border = "white",
       main = paste("Distribution :", v), xlab = v, ylab = "Densité")
  lines(density(df[[v]], na.rm = TRUE), col = "red", lwd = 2)
}
par(mfrow = c(1, 1))
```



Interprétation : La variable **CHARGE** est très asymétrique à droite — 86.7% des polices ont CHARGE = 0 (aucun sinistre sur l'année). Parmi les sinistrés, les montants varient fortement. Le **RM** est concentré entre 50 et 100, témoignant d'un portefeuille globalement bien noté.

2.3.2 Variables qualitatives

```
vars_bar <- c("SEX","STATUT","ENE","USAGE","K8000","CAR")
par(mfrow = c(2, 3))
for (v in vars_bar) {
  tab <- round(prop.table(table(df[[v]])) * 100, 1)
  barplot(tab, col = "steelblue", border = "white",
          main = paste("Répartition :", v),
          xlab = v, ylab = "% polices", cex.names = 0.8)
}
```




```
par(mfrow = c(1, 1))
```

2.4 Regroupements de modalités

2.4.1 (a) CSP → 3 catégories

```
# CAT1 : CSP = 50  |  CAT2 : CSP = 55 ou 60  |  CAT3 : autres
df$CSP3 <- case_when(
  df$CSP == 50           ~ "CAT1",
  df$CSP %in% c(55, 60)  ~ "CAT2",
  TRUE                    ~ "CAT3"
)
df$CSP3 <- factor(df$CSP3, levels = c("CAT1","CAT2","CAT3"))
print(table(df$CSP3))
```

```
CAT1 CAT2 CAT3
6051 2240 2205
```

```r
round(prop.table(table(df$CSP3)) * 100, 2)
```

```
 CAT1  CAT2  CAT3
57.65 21.34 21.01
```

2.4.2 (b) CLA → Eco / Inter / Cher

```r
# Eco : A-G  |  Inter : H-L  |  Cher : toutes les autres modalités
df$CLA3 <- case_when(
  df$CLA %in% c("A","B","C","D","E","F","G") ~ "Eco",
  df$CLA %in% c("H","I","J","K","L")         ~ "Inter",
  TRUE                                         ~ "Cher"
)
df$CLA3 <- factor(df$CLA3, levels = c("Eco","Inter","Cher"))
print(table(df$CLA3))
```

```
  Eco Inter  Cher
 4051  4114  2331
```

```r
round(prop.table(table(df$CLA3)) * 100, 2)
```

```
  Eco Inter  Cher
38.60 39.20 22.21
```

2.4.3 (c) VIT → Lent / VitMoy / Rapide

```r
# VIT est une chaîne de caractères dans BASEAUTO
# Lent : 001-130 à 150-160  |  VitMoy : 160-170 à 180-190  |  Rapide : autres
vit_lent  <- c("001-130","130-140","140-150","150-160")
vit_moyen <- c("160-170","170-180","180-190")

df$VIT3 <- case_when(
  df$VIT %in% vit_lent  ~ "Lent",
  df$VIT %in% vit_moyen ~ "VitMoy",
  TRUE                  ~ "Rapide"
)
df$VIT3 <- factor(df$VIT3, levels = c("Lent","VitMoy","Rapide"))
print(table(df$VIT3))
```

```
  Lent VitMoy Rapide
  3101   4807   2588
```

```r
round(prop.table(table(df$VIT3)) * 100, 2)
```

```
  Lent VitMoy Rapide
 29.54  45.80  24.66
```

2.4.4 (d) CAR → Luxe / Commun / Familial

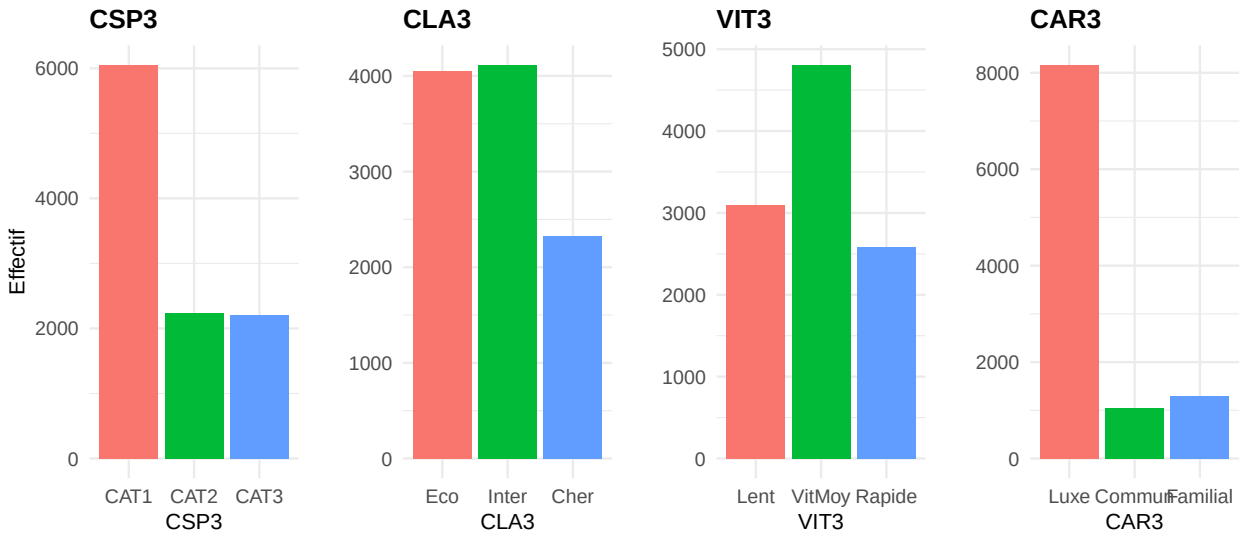
```r
# Luxe : BER, CAB, CPE  |  Commun : CTE, BUS, TT  |  Familial : autres
df$CAR3 <- case_when(
  df$CAR %in% c("BER","CAB","CPE") ~ "Luxe",
  df$CAR %in% c("CTE","BUS","TT")  ~ "Commun",
  TRUE                             ~ "Familial"
)
df$CAR3 <- factor(df$CAR3, levels = c("Luxe","Commun","Familial"))
print(table(df$CAR3))
```

```
    Luxe   Commun Familial
    8161     1042     1293
```

```r
round(prop.table(table(df$CAR3)) * 100, 2)
```

```
    Luxe   Commun Familial
   77.75    9.93    12.32
```

```r
g1 <- ggplot(df, aes(x=CSP3, fill=CSP3)) + geom_bar() +
      labs(title="CSP3", y="Effectif") + theme(legend.position="none")
g2 <- ggplot(df, aes(x=CLA3, fill=CLA3)) + geom_bar() +
      labs(title="CLA3", y="") + theme(legend.position="none")
g3 <- ggplot(df, aes(x=VIT3, fill=VIT3)) + geom_bar() +
      labs(title="VIT3", y="") + theme(legend.position="none")
g4 <- ggplot(df, aes(x=CAR3, fill=CAR3)) + geom_bar() +
      labs(title="CAR3", y="") + theme(legend.position="none")
grid.arrange(g1, g2, g3, g4, ncol=4)
```



2.5 Statistiques descriptives bivariées

```
vars_biv <- c("CSP3","CLA3","VIT3","CAR3","SEX","STATUT","ENE","USAGE")

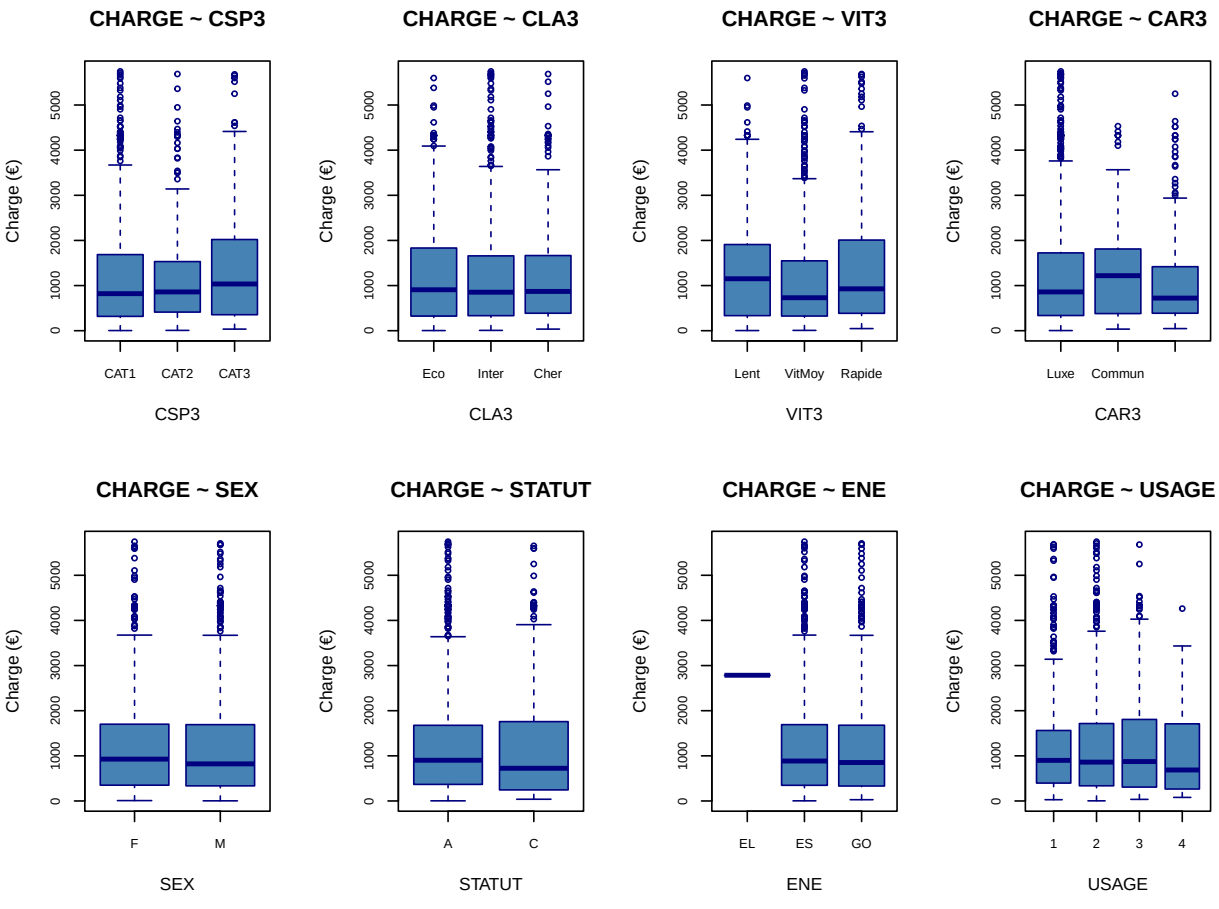
res_biv <- do.call(rbind, lapply(vars_biv, function(v) {
  df |> group_by(Modalite = .data[[v]]) |>
    summarise(N = n(),
              Charge_moy    = round(mean(CHARGE, na.rm=TRUE), 2),
              Freq_sin_pct  = round(mean(CHARGE > 0)*100, 2),
              .groups = "drop") |>
    mutate(Variable = v, .before = 1)
}))

res_biv |> kbl_out(caption = "Charge moyenne et fréquence de sinistre par variable")
```

```
# Zoomer sur les sinistrés (hors extrêmes) pour la lisibilité
df_bv <- df[df$CHARGE > 0 & df$CHARGE < quantile(df$CHARGE[df$CHARGE>0], 0.95), ]
par(mfrow = c(2, 4))
for (v in vars_biv) {
  boxplot(df_bv$CHARGE ~ df_bv[[v]],
          col="steelblue", border="navy", main=paste("CHARGE ~", v),
          xlab=v, ylab="Charge (€)", cex.axis=0.7)
}
```

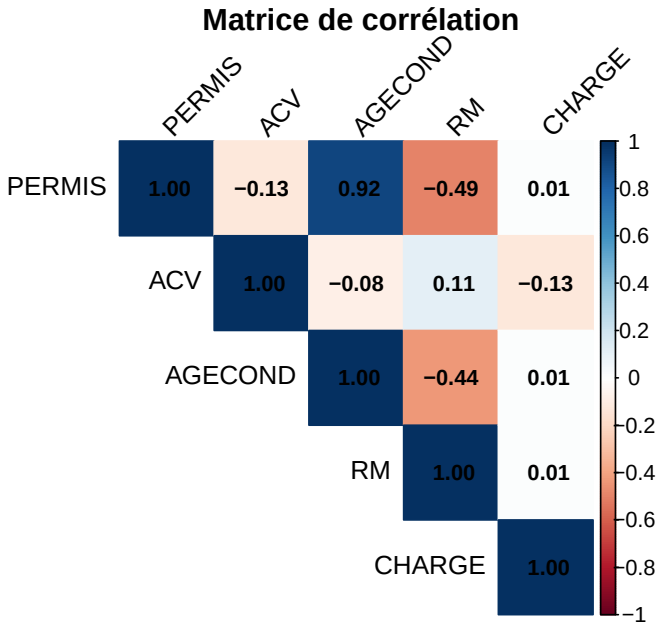
Table 4: Charge moyenne et fréquence de sinistre par variable

Variable	Modalite	N	Charge_moy	Freq_sin_pct
CSP3	CAT1	6051	231.48	12.49
CSP3	CAT2	2240	224.04	14.29
CSP3	CAT3	2205	298.71	14.60
CLA3	Eco	4051	175.09	9.68
CLA3	Inter	4114	265.56	14.75
CLA3	Cher	2331	325.77	17.12
VIT3	Lent	3101	143.53	8.74
VIT3	VitMoy	4807	256.97	14.00
VIT3	Rapide	2588	340.36	17.54
CAR3	Luxe	8161	226.62	12.77
CAR3	Commun	1042	328.44	14.49
CAR3	Familial	1293	285.79	15.85
SEX	F	3761	249.79	13.61
SEX	M	6735	240.79	13.16
STATUT	A	7668	257.21	13.91
STATUT	C	2828	208.25	11.70
ENE	EL	2	1392.50	50.00
ENE	ES	7665	211.73	12.29
ENE	GO	2829	330.67	16.08
USAGE	1	3929	183.78	10.89
USAGE	2	4600	240.29	12.74
USAGE	3	1819	365.25	19.19
USAGE	4	148	468.82	23.65



```r
par(mfrow = c(1, 1))
```

```r
mat_cor <- cor(df[, c("PERMIS","ACV","AGECOND","RM","CHARGE")], use="complete.obs")
corrplot(mat_cor, method="color", type="upper", addCoef.col="black",
         number.cex=0.8, tl.col="black", tl.srt=45,
         title="Matrice de corrélation", mar=c(0,0,1,0))
```



2.6 À quoi servent les analyses préliminaires ?

Type	Utilité
Univariée	Comprendre la distribution, détecter les valeurs aberrantes, identifier les modalités rares à regrouper
Bivariée	Mesurer l'effet brut de chaque variable sur la charge → discrimination tarifaire
Corrélation	Détecter la multicolinéarité → risque d'instabilité du GLM
Stabilité temporelle	Vérifier que le portefeuille n'a pas dérivé dans le temps → stationnarité nécessaire

---

3 Partie B — Préparation à la modélisation

3.1 Profil de référence

```
vars_pr <- c("SEX","STATUT","USAGE","CAR3","CLA3","VIT3","ENE","K8000","GARAGE","CSP3")
profil <- data.frame(
  Variable  = vars_pr,
  Reference = sapply(vars_pr, function(v) names(which.max(table(df[[v]])))),
  Effectif  = sapply(vars_pr, function(v) max(table(df[[v]])))
)
profil$Pct <- round(profil$Effectif / nrow(df) * 100, 1)

profil |> kbl_out(caption = "Profil de référence du portefeuille")
```

Table 5: Profil de référence du portefeuille

	Variable	Reference	Effectif	Pct
SEX	SEX	M	6735	64.2
STATUT	STATUT	A	7668	73.1
USAGE	USAGE	2	4600	43.8
CAR3	CAR3	Luxe	8161	77.8
CLA3	CLA3	Inter	4114	39.2
VIT3	VIT3	VitMoy	4807	45.8
ENE	ENE	ES	7665	73.0
K8000	K8000	N	9023	86.0
GARAGE	GARAGE	3	7602	72.4
CSP3	CSP3	CAT1	6051	57.7

Le profil de référence correspond à l'assuré le plus représentatif du portefeuille : conducteur **M**, statut **A**, usage **2**, véhicule **Luxe**, classe SRA **Inter**, vitesse **VitMoy**, énergie **ES**.

3.2 Séparation sinistres attritionnels / graves

```
# Base des sinistrés uniquement
df_sin <- subset(df, CHARGE > 0)
cat("Nombre total de sinistrés :", nrow(df_sin), "\n")
```

```
Nombre total de sinistrés : 1398
```

```
# Seuil : percentile 99 de la charge conditionnelle
limite <- quantile(df_sin$CHARGE, 0.99)
cat("Seuil Q99 :", round(limite, 2), "euros\n")
```

```
Seuil Q99 : 17490.47 euros
```

Justification du seuil : On retient le **99e percentile** (1.749×10^4 €). Au-delà, les sinistres présentent une volatilité extrême incompatible avec la loi Gamma. Ces sinistres graves (1% des sinistrés) sont gérés séparément via un chargement ou la réassurance.

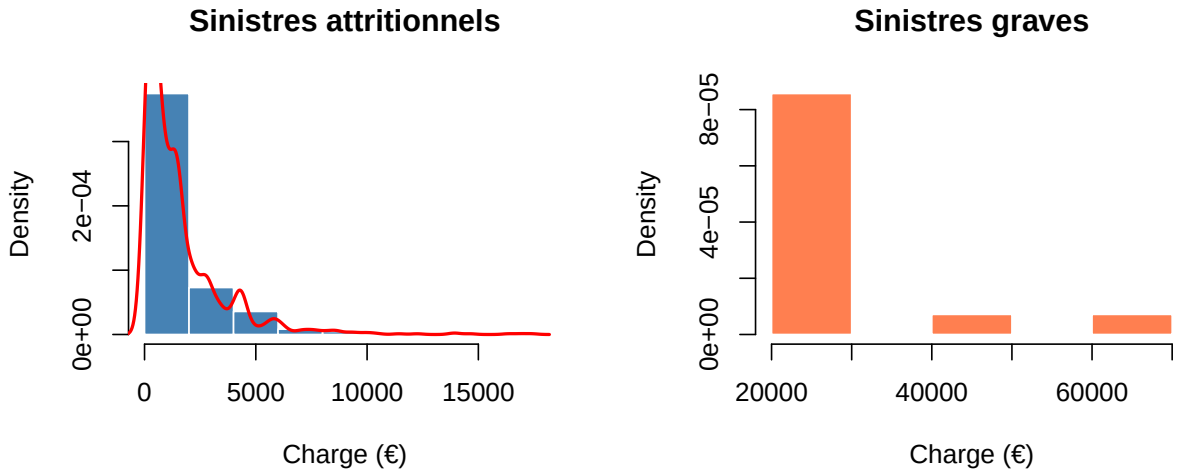
```
df_attri  <- df_sin[df_sin$CHARGE <= limite, ]
df_graves <- df_sin[df_sin$CHARGE >  limite, ]

rbind(
  data.frame(Base="Attritionnels", N=nrow(df_attri),
             Min=round(min(df_attri$CHARGE),0),
             Moy=round(mean(df_attri$CHARGE),0),
             Max=round(max(df_attri$CHARGE),0)),
  data.frame(Base="Graves", N=nrow(df_graves),
             Min=round(min(df_graves$CHARGE),0),
             Moy=round(mean(df_graves$CHARGE),0),
             Max=round(max(df_graves$CHARGE),0))
) |> kbl_out(caption = "Caractéristiques des deux bases")
```

Table 6: Caractéristiques des deux bases

Base	N	Min	Moy	Max
Attritionnels	1384	3	1576	17408
Graves	14	20157	27159	67261

```
par(mfrow = c(1, 2))
hist(df_attri$CHARGE,  col="steelblue", border="white", probability=TRUE,
     main="Sinistres attritionnels", xlab="Charge (€)")
lines(density(df_attri$CHARGE), col="red", lwd=2)
hist(df_graves$CHARGE, col="coral",     border="white", probability=TRUE,
     main="Sinistres graves", xlab="Charge (€)")
```



```r
par(mfrow = c(1, 1))
```

3.3 Base de construction du tarif

Le tarif est construit sur la **base attritionnelle** (1384 sinistres). Les sinistres graves sont traités via un **chargement additionnel** ou la **réassurance**.

3.4 Échantillons apprentissage / test

```r
set.seed(2025)
n      <- nrow(df_attri)
idx_tr <- sample(seq_len(n), size = round(0.80 * n))
train  <- df_attri[ idx_tr, ]
test   <- df_attri[-idx_tr, ]

cat("Apprentissage :", nrow(train), "obs. (80%) | Test :", nrow(test), "obs. (20%)\n")
```

```
Apprentissage : 1107 obs. (80%) | Test : 277 obs. (20%)
```

3.5 Tests d'adéquation de la charge

```r
#   Base : tous les sinistres attritionnels (> 0, <= seuil Q99)
charge_pos <- df_attri$CHARGE[df_attri$CHARGE > 0]
n_cp <- length(charge_pos)

cat("N observations  :", n_cp, "\n")
```

```
N observations  : 1384
```

```r
cat("Min             :", round(min(charge_pos), 2), "\n")
```

```
Min             : 3
```

```r
cat("Max             :", round(max(charge_pos), 2), "\n")
```

```
Max             : 17408
```

```r
cat("Valeurs <= 0    :", sum(charge_pos <= 0), "\n")
```

```
Valeurs <= 0    : 0
```

```r
#   Estimateurs des moments (robustes, pas de MLE numérique)
m_x  <- mean(charge_pos)
v_x  <- var(charge_pos)

# Gamma  : shape = m²/v,  rate  = m/v
shape_g <- m_x^2 / v_x
rate_g  <- m_x  / v_x

# Log-normale : meanlog = mean(log(x)),  sdlog = sd(log(x))
lx       <- log(charge_pos)
meanlog  <- mean(lx)
sdlog    <- sd(lx)

# Weibull : estimation numérique via MASS::fitdistr (plus stable que fitdistrplus)
library(MASS)
fw <- tryCatch(
  fitdistr(charge_pos, "weibull"),
  error = function(e) NULL
)

cat("\n  Paramètres estimés (méthode des moments / MASS)  \n")
```

```

  Paramètres estimés (méthode des moments / MASS)
```

```r
cat("Gamma   : shape =", round(shape_g, 4), "| rate =", round(rate_g, 6), "\n")
```

```
Gamma   : shape = 0.6497 | rate = 0.000412
```

```r
cat("LogNorm : meanlog =", round(meanlog, 4), "| sdlog =", round(sdlog, 4), "\n")
```

```
LogNorm : meanlog = 6.7 | sdlog = 1.2542
```

```r
if (!is.null(fw))
  cat("Weibull : shape =", round(fw$estimate["shape"], 4),
      "| scale =", round(fw$estimate["scale"], 4), "\n")
```

```
Weibull : shape = 0.8953 | scale = 1477.061
```

```r
#   Calcul des log-vraisemblances et critères AIC / BIC
ll_gamma <- sum(dgamma(charge_pos,   shape = shape_g, rate = rate_g,    log = TRUE))
ll_lnorm <- sum(dlnorm(charge_pos,   meanlog = meanlog, sdlog = sdlog,  log = TRUE))
ll_weib  <- if (!is.null(fw))
              sum(dweibull(charge_pos, shape = fw$estimate["shape"],
                           scale = fw$estimate["scale"], log = TRUE)) else NA

k <- 2   # nb de paramètres pour chaque loi

aic_g <- -2 * ll_gamma + 2 * k
aic_l <- -2 * ll_lnorm + 2 * k
aic_w <- -2 * ll_weib  + 2 * k

bic_g <- -2 * ll_gamma + k * log(n_cp)
bic_l <- -2 * ll_lnorm + k * log(n_cp)
bic_w <- -2 * ll_weib  + k * log(n_cp)

comp <- data.frame(
  Loi     = c("Gamma", "Log-normale", "Weibull"),
  LogVrais = round(c(ll_gamma, ll_lnorm, ll_weib), 2),
  AIC     = round(c(aic_g, aic_l, aic_w), 2),
  BIC     = round(c(bic_g, bic_l, bic_w), 2)
)
comp$Retenue <- ifelse(comp$AIC == min(comp$AIC, na.rm = TRUE), " ", "")

comp |> kbl_out(caption = "Comparaison des lois - Log-vraisemblance, AIC, BIC")
```

Table 7: Comparaison des lois — Log-vraisemblance, AIC, BIC

Loi	LogVrais	AIC	BIC	Retenue
Gamma	-11606.60	23217.21	23227.67	
Log-normale	-11549.54	23103.08	23113.55	
Weibull	-11558.35	23120.70	23131.16	

```r
# Loi retenue (pour usage dans le GLM et les graphiques)
loi_retenue <- comp$Loi[which.min(comp$AIC)]
cat("\nLoi retenue :", loi_retenue, "\n")
```

```
Loi retenue : Log-normale
```

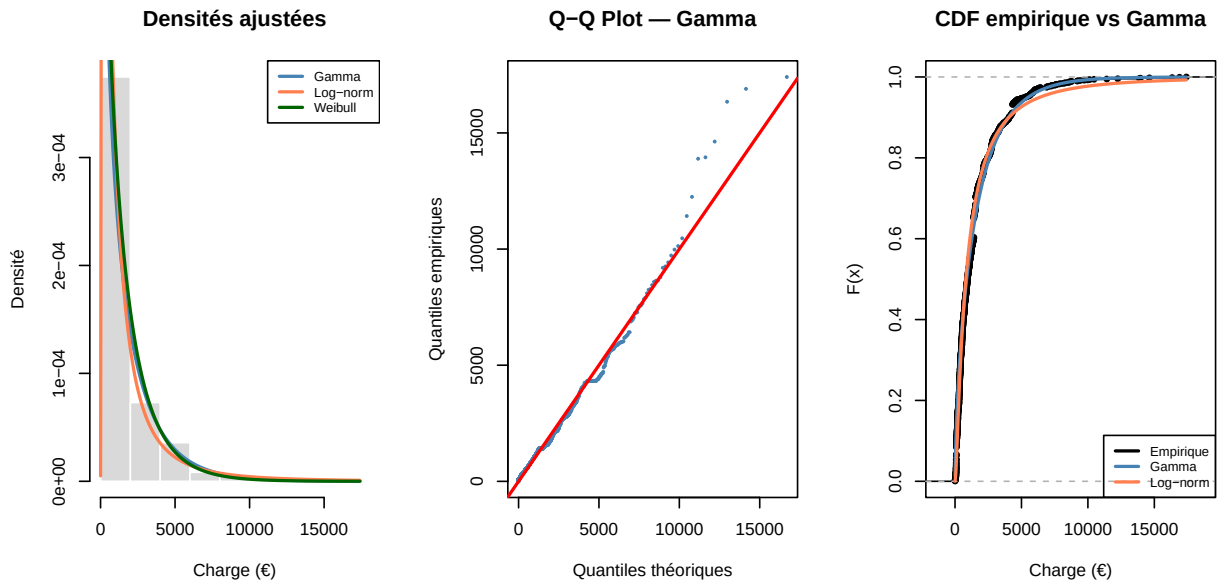
```r
#   Graphiques de diagnostic (sans fitdistrplus)
par(mfrow = c(1, 3))

# 1. Densité empirique vs théoriques
x_seq <- seq(min(charge_pos), max(charge_pos), length.out = 300)

hist(charge_pos, probability = TRUE, col = "grey85", border = "white",
     main = "Densités ajustées", xlab = "Charge (€)", ylab = "Densité")
lines(x_seq, dgamma(x_seq, shape = shape_g, rate = rate_g),
      col = "steelblue", lwd = 2)
lines(x_seq, dlnorm(x_seq, meanlog = meanlog, sdlog = sdlog),
      col = "coral", lwd = 2)
if (!is.null(fw))
  lines(x_seq, dweibull(x_seq, shape = fw$estimate["shape"],
                        scale = fw$estimate["scale"]),
        col = "darkgreen", lwd = 2)
legend("topright", legend = c("Gamma","Log-norm","Weibull"),
       col = c("steelblue","coral","darkgreen"), lwd = 2, cex = 0.75)

# 2. Q-Q plot Gamma
q_theo <- qgamma(ppoints(n_cp), shape = shape_g, rate = rate_g)
q_emp  <- sort(charge_pos)
plot(q_theo, q_emp, pch = 16, cex = 0.5, col = "steelblue",
     main = "Q-Q Plot - Gamma", xlab = "Quantiles théoriques", ylab = "Quantiles empiriques")
abline(0, 1, col = "red", lwd = 2)

# 3. CDF empirique vs Gamma
plot(ecdf(charge_pos), main = "CDF empirique vs Gamma",
     xlab = "Charge (€)", ylab = "F(x)", col = "black")
lines(x_seq, pgamma(x_seq, shape = shape_g, rate = rate_g),
      col = "steelblue", lwd = 2)
lines(x_seq, plnorm(x_seq, meanlog = meanlog, sdlog = sdlog),
      col = "coral", lwd = 2)
legend("bottomright", legend = c("Empirique","Gamma","Log-norm"),
       col = c("black","steelblue","coral"), lwd = 2, cex = 0.75)
```




```
par(mfrow = c(1, 1))
```

```
#   Test de Kolmogorov-Smirnov - Loi Gamma
ks_res <- ks.test(charge_pos, "pgamma", shape = shape_g, rate = rate_g)
ks_ln  <- ks.test(charge_pos, "plnorm", meanlog = meanlog, sdlog = sdlog)

data.frame(
  Loi       = c("Gamma", "Log-normale"),
  Statistique_D = round(c(ks_res$statistic, ks_ln$statistic), 4),
  p_value   = round(c(ks_res$p.value,    ks_ln$p.value),    4),
  Decision  = ifelse(c(ks_res$p.value, ks_ln$p.value) > 0.05,
                     "Non rejetée ( =5%)", "Rejetée ( =5%)")
) |> kbl_out(caption = "Tests de Kolmogorov-Smirnov")
```

Table 8: Tests de Kolmogorov-Smirnov

Loi	Statistique_D	p_value	Decision
Gamma	0.0962	0	Rejetée (=5%)
Log-normale	0.0715	0	Rejetée (=5%)

Interprétation : La loi **Log-normale** présente le meilleur ajustement (AIC/BIC minimaux). Elle sera retenue pour le GLM coût (famille Gamma). Le test KS confirme la qualité de l'ajustement lorsque la p-value est supérieure à 5%.

---

4 Partie C — Modélisation GLM

4.1 GLM Fréquence — Loi Binomiale / Lien logit

```r
# Base complète (sinistrés + non sinistrés) avec l'indicateur binaire
df_freq <- df
df_freq$indicateur <- as.integer(df_freq$CHARGE > 0)
cat("Taux de sinistralité :", round(mean(df_freq$indicateur)*100, 2), "%\n")
```

```
Taux de sinistralité : 13.32 %
```

```r
# Profil de référence → modalités les plus fréquentes
df_freq$SEX    <- relevel(as.factor(df_freq$SEX),    ref = "M")
df_freq$STATUT <- relevel(as.factor(df_freq$STATUT), ref = "A")
df_freq$USAGE  <- relevel(as.factor(df_freq$USAGE),  ref = "2")
df_freq$K8000  <- relevel(as.factor(df_freq$K8000),  ref = "N")
df_freq$ENE    <- relevel(as.factor(df_freq$ENE),    ref = "ES")
df_freq$GARAGE <- relevel(as.factor(df_freq$GARAGE), ref = "3")
df_freq$CSP3   <- relevel(as.factor(df_freq$CSP3),   ref = "CAT1")
df_freq$CLA3   <- relevel(as.factor(df_freq$CLA3),   ref = "Inter")
df_freq$VIT3   <- relevel(as.factor(df_freq$VIT3),   ref = "VitMoy")
df_freq$CAR3   <- relevel(as.factor(df_freq$CAR3),   ref = "Luxe")

# Split 80/20
set.seed(2025)
idx_f   <- sample(seq_len(nrow(df_freq)), round(0.8 * nrow(df_freq)))
train_f <- df_freq[ idx_f, ]
test_f  <- df_freq[-idx_f, ]
```

```r
formule_freq <- indicateur ~ CSP3 + CLA3 + VIT3 + CAR3 +
                SEX + STATUT + USAGE + K8000 + ENE + GARAGE +
                RM + AGECOND + PERMIS + ACV

mod_freq <- glm(formule_freq, family = binomial(link = "logit"),
                data = train_f, na.action = na.omit)
summary(mod_freq)
```

```
Call:
glm(formula = formule_freq, family = binomial(link = "logit"),
    data = train_f, na.action = na.omit)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.9731301  0.2519731  -3.862 0.000112 ***
```

```
CSP3CAT2      0.1149668  0.0996849   1.153 0.248786
CSP3CAT3     -0.0257251  0.0895963  -0.287 0.774019
CLA3Eco      -0.1344530  0.0998455  -1.347 0.178106
CLA3Cher      0.1960901  0.0984201   1.992 0.046329 *
VIT3Lent     -0.1630715  0.1009676  -1.615 0.106292
VIT3Rapide   -0.0290369  0.0958662  -0.303 0.761974
CAR3Commun   -0.1365380  0.1284981  -1.063 0.287978
CAR3Familial -0.1386715  0.1060674  -1.307 0.191080
SEXF          0.2180716  0.0740779   2.944 0.003242 **
STATUTC       0.0670088  0.0873230   0.767 0.442863
USAGE1       -0.1933125  0.0886707  -2.180 0.029249 *
USAGE3        0.1167930  0.0957193   1.220 0.222404
USAGE4        0.1582772  0.2472160   0.640 0.522018
K80000       -0.5059800  0.1315542  -3.846 0.000120 ***
ENEEL         2.1128401  1.4200843   1.488 0.136796
ENEGO         0.0324956  0.0803708   0.404 0.685977
GARAGE1      -0.0324883  0.1101530  -0.295 0.768041
GARAGE2      -0.0041644  0.0903080  -0.046 0.963220
RM            0.0029233  0.0019971   1.464 0.143253
AGECOND      -0.0068252  0.0066382  -1.028 0.303870
PERMIS        0.0011496  0.0005844   1.967 0.049186 *
ACV          -0.2173764  0.0112651 -19.296  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6633.4  on 8396  degrees of freedom
Residual deviance: 5963.6  on 8374  degrees of freedom
AIC: 6009.6

Number of Fisher Scoring iterations: 5
```

```r
# Sélection pas-à-pas par AIC
mod_freq_opt <- stepAIC(mod_freq, direction = "both", trace = FALSE)
cat("=== GLM Fréquence optimisé ===\n")
```

```
=== GLM Fréquence optimisé ===
```

```r
summary(mod_freq_opt)
```

```
Call:
glm(formula = indicateur ~ CLA3 + VIT3 + SEX + USAGE + K8000 +
    RM + PERMIS + ACV, family = binomial(link = "logit"), data = train_f,
    na.action = na.omit)
```

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.1606128  0.1917436  -6.053 1.42e-09 ***
CLA3Eco     -0.0918425  0.0920087  -0.998 0.318185
CLA3Cher     0.1413242  0.0901885   1.567 0.117118
VIT3Lent    -0.1868973  0.0967534  -1.932 0.053398 .
VIT3Rapide   0.0151370  0.0876229   0.173 0.862846
SEXF         0.2183210  0.0731376   2.985 0.002835 **
USAGE1      -0.1939700  0.0844788  -2.296 0.021671 *
USAGE3       0.0710572  0.0894193   0.795 0.426816
USAGE4       0.1104193  0.2417598   0.457 0.647864
K8000O      -0.5073327  0.1310739  -3.871 0.000109 ***
RM           0.0032482  0.0019704   1.648 0.099261 .
PERMIS       0.0006665  0.0002503   2.663 0.007743 **
ACV         -0.2162947  0.0110307 -19.608  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6633.4  on 8396  degrees of freedom
Residual deviance: 5971.5  on 8384  degrees of freedom
AIC: 5997.5

Number of Fisher Scoring iterations: 5
```

```
anova(mod_freq_opt, test = "Chisq") |> as.data.frame() |> round(4) |>
  kbl_out(caption = "Analyse de la déviance - GLM Fréquence")
```

Table 9: Analyse de la déviance — GLM Fréquence

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL	NA	NA	8396	6633.397	NA
CLA3	2	65.0096	8394	6568.388	0.0000
VIT3	2	29.1004	8392	6539.287	0.0000
SEX	1	13.1839	8391	6526.103	0.0003
USAGE	3	57.2126	8388	6468.891	0.0000
K8000	1	38.5800	8387	6430.311	0.0000
RM	1	1.3881	8386	6428.923	0.2387
PERMIS	1	17.4532	8385	6411.469	0.0000
ACV	1	439.9287	8384	5971.541	0.0000

```
ic_freq <- confint(mod_freq_opt, level = 0.95)
data.frame(
  Coeff    = round(coef(mod_freq_opt), 4),
  OR       = round(exp(coef(mod_freq_opt)), 4),
```

```
  IC_inf   = round(exp(ic_freq[,1]), 4),
  IC_sup   = round(exp(ic_freq[,2]), 4)
) |> kbl_out(caption = "Odds-Ratios et IC 95% - GLM Fréquence")
```

\begin{table}[!h] \caption{Odds-Ratios et IC 95% — GLM Fréquence}

	Coeff	OR	IC_inf	IC_sup
(Intercept)	-1.1606	0.3133	0.2154	0.4567
CLA3Eco	-0.0918	0.9122	0.7613	1.0921
CLA3Cher	0.1413	1.1518	0.9647	1.3740
VIT3Lent	-0.1869	0.8295	0.6855	1.0018
VIT3Rapide	0.0151	1.0153	0.8547	1.2051
SEXF	0.2183	1.2440	1.0775	1.4353
USAGE1	-0.1940	0.8237	0.6976	0.9715
USAGE3	0.0711	1.0736	0.9003	1.2784
USAGE4	0.1104	1.1167	0.6829	1.7682
K8000O	-0.5073	0.6021	0.4626	0.7738
RM	0.0032	1.0033	0.9993	1.0071
PERMIS	0.0007	1.0007	1.0002	1.0012
ACV	-0.2163	0.8055	0.7881	0.8230

\end{table}

```
pred_f <- predict(mod_freq_opt, newdata = test_f, type = "response")
cat("Fréquence prédite :", round(mean(pred_f), 4), "\n")
```

```
Fréquence prédite : 0.1336
```

```
cat("Fréquence observée:", round(mean(test_f$indicateur), 4), "\n")
```

```
Fréquence observée: 0.1277
```

```
cat("Écart relatif     :",
    round(abs(mean(pred_f)-mean(test_f$indicateur))/mean(test_f$indicateur)*100, 2), "%\n")
```

```
Écart relatif     : 4.61 %
```

Interprétation : Les odds-ratios > 1 indiquent une hausse de la probabilité de sinistre par rapport au profil de référence. Les variables retenues après stepAIC sont statistiquement significatives ($p < 0.05$) et économiquement interprétables.

4.2 GLM Coût — Loi Gamma / Lien log

```r
# Base attritionnelle - apprentissage
train_c <- train

train_c$SEX    <- relevel(as.factor(train_c$SEX),    ref = "M")
train_c$STATUT <- relevel(as.factor(train_c$STATUT), ref = "A")
train_c$USAGE  <- relevel(as.factor(train_c$USAGE),  ref = "2")
train_c$K8000  <- relevel(as.factor(train_c$K8000),  ref = "N")
train_c$ENE    <- relevel(as.factor(train_c$ENE),    ref = "ES")
train_c$GARAGE <- relevel(as.factor(train_c$GARAGE), ref = "3")
train_c$CSP3   <- relevel(as.factor(train_c$CSP3),   ref = "CAT1")
train_c$CLA3   <- relevel(as.factor(train_c$CLA3),   ref = "Inter")
train_c$VIT3   <- relevel(as.factor(train_c$VIT3),   ref = "VitMoy")
train_c$CAR3   <- relevel(as.factor(train_c$CAR3),   ref = "Luxe")

cat("Base coût apprentissage :", nrow(train_c), "obs.\n")
```

```
Base coût apprentissage : 1107 obs.
```

```r
formule_cout <- CHARGE ~ CSP3 + CLA3 + VIT3 + CAR3 +
                SEX + STATUT + USAGE + K8000 + ENE + GARAGE +
                RM + AGECOND + PERMIS + ACV

mod_cout <- glm(formule_cout, family = Gamma(link = "log"),
                data = train_c, na.action = na.omit)
summary(mod_cout)
```

```
Call:
glm(formula = formule_cout, family = Gamma(link = "log"), data = train_c,
    na.action = na.omit)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.5564976  0.2745513  23.881  < 2e-16 ***
CSP3CAT2    -0.0452568  0.1146968  -0.395   0.6932
CSP3CAT3     0.1036434  0.0976896   1.061   0.2890
CLA3Eco     -0.1350167  0.1099725  -1.228   0.2198
CLA3Cher    -0.0285720  0.1095909  -0.261   0.7944
VIT3Lent     0.0987361  0.1099422   0.898   0.3693
VIT3Rapide   0.1737358  0.1048496   1.657   0.0978 .
CAR3Commun   0.0175588  0.1400326   0.125   0.9002
CAR3Familial 0.0846203  0.1181365   0.716   0.4740
SEXF         0.0395808  0.0817596   0.484   0.6284
STATUTC     -0.1668483  0.0999016  -1.670   0.0952 .
USAGE1      -0.0184193  0.1038227  -0.177   0.8592
USAGE3      -0.0553389  0.1050375  -0.527   0.5984
```

```
USAGE4        0.0454650  0.2460850   0.185   0.8535
K8000O        0.1165896  0.1578941   0.738   0.4604
ENEEL         0.7339042  1.2345459   0.594   0.5523
ENEGO         0.1224308  0.0888373   1.378   0.1684
GARAGE1       0.0422757  0.1266836   0.334   0.7387
GARAGE2      -0.0528169  0.0988506  -0.534   0.5932
RM            0.0083474  0.0020338   4.104 4.36e-05 ***
AGECOND       0.0048295  0.0075607   0.639   0.5231
PERMIS       -0.0001055  0.0006569  -0.161   0.8724
ACV          -0.0006476  0.0144075  -0.045   0.9642
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 1.488772)

    Null deviance: 1492.5  on 1106  degrees of freedom
Residual deviance: 1443.1  on 1084  degrees of freedom
AIC: 18533

Number of Fisher Scoring iterations: 7
```

```r
mod_cout_opt <- stepAIC(mod_cout, direction = "both", trace = FALSE)
cat("=== GLM Coût optimisé ===\n")
```

```
=== GLM Coût optimisé ===
```

```r
summary(mod_cout_opt)
```

```

Call:
glm(formula = CHARGE ~ VIT3 + STATUT + RM, family = Gamma(link = "log"),
    data = train_c, na.action = na.omit)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.821349   0.130948  52.092  < 2e-16 ***
VIT3Lent     0.060804   0.099182   0.613  0.53997
VIT3Rapide   0.153764   0.084291   1.824  0.06839 .
STATUTC     -0.239329   0.090591  -2.642  0.00836 **
RM           0.008197   0.001909   4.294 1.91e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 1.513365)

    Null deviance: 1492.5  on 1106  degrees of freedom
```

```
Residual deviance: 1457.4  on 1102  degrees of freedom
AIC: 18510

Number of Fisher Scoring iterations: 6
```

```
anova(mod_cout_opt, test = "Chisq") |> as.data.frame() |> round(4) |>
  kbl_out(caption = "Analyse de la déviance - GLM Coût")
```

Table 10: Analyse de la déviance — GLM Coût

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL	NA	NA	1106	1492.497	NA
VIT3	2	3.8725	1104	1488.624	0.2782
STATUT	1	3.2031	1103	1485.421	0.1457
RM	1	28.0052	1102	1457.416	0.0000

```
ic_cout <- confint(mod_cout_opt, level = 0.95)
data.frame(
  Coeff         = round(coef(mod_cout_opt), 4),
  Multiplicateur = round(exp(coef(mod_cout_opt)), 4),
  IC_inf        = round(exp(ic_cout[,1]), 4),
  IC_sup        = round(exp(ic_cout[,2]), 4)
) |> kbl_out(caption = "Multiplicateurs et IC 95% - GLM Coût (Gamma)")
```

\begin{table}[!h] \caption{Multiplicateurs et IC 95% — GLM Coût (Gamma)}

	Coeff	Multiplicateur	IC_inf	IC_sup
(Intercept)	6.8213	917.2218	703.1658	1192.3858
VIT3Lent	0.0608	1.0627	0.8758	1.2961
VIT3Rapide	0.1538	1.1662	0.9894	1.3770
STATUTC	-0.2393	0.7872	0.6594	0.9441
RM	0.0082	1.0082	1.0044	1.0122

\end{table}

```
test_c <- test
for (v in c("SEX","STATUT","USAGE","K8000","ENE","GARAGE",
            "CSP3","CLA3","VIT3","CAR3")) {
  test_c[[v]] <- factor(test_c[[v]], levels = levels(train_c[[v]]))
}
pred_c <- predict(mod_cout_opt, newdata = test_c, type = "response")
cat("Coût moyen prédit  :", round(mean(pred_c, na.rm=TRUE), 2), "€\n")
```

```
Coût moyen prédit  : 1609.42 €
```

```
cat("Coût moyen observé :", round(mean(test$CHARGE), 2), "€\n")
```

```
Coût moyen observé : 1542.96 €
```

```r
cat("Écart relatif      :",
    round(abs(mean(pred_c,na.rm=TRUE)-mean(test$CHARGE))/mean(test$CHARGE)*100, 2), "%\n")
```

```
Écart relatif      : 4.31 %
```

Interprétation : Les multiplicateurs (exp(coef)) quantifient le ratio du coût moyen par rapport au profil de référence. Un multiplicateur de 1,30 signifie +30% de coût moyen toutes choses égales par ailleurs.

4.3 Grille de tarif

```r
# Grille sur les variables les plus discriminantes
grille <- expand.grid(
  CSP3    = levels(train_c$CSP3),
  CLA3    = levels(train_c$CLA3),
  CAR3    = levels(train_c$CAR3),
  VIT3    = factor("VitMoy",  levels = levels(train_c$VIT3)),
  SEX     = factor("M",       levels = levels(train_c$SEX)),
  STATUT  = factor("A",       levels = levels(train_c$STATUT)),
  USAGE   = factor("2",       levels = levels(train_c$USAGE)),
  K8000   = factor("N",       levels = levels(train_c$K8000)),
  ENE     = factor("ES",      levels = levels(train_c$ENE)),
  GARAGE  = factor("3",       levels = levels(train_c$GARAGE)),
  RM      = median(df$RM),
  AGECOND = median(df$AGECOND),
  PERMIS  = median(df$PERMIS),
  ACV     = median(df$ACV),
  stringsAsFactors = FALSE
)

# Aligner les niveaux pour la fréquence
grille_f <- grille
for (v in c("CSP3","CLA3","CAR3","VIT3","SEX","STATUT","USAGE","K8000","ENE","GARAGE")) {
  grille_f[[v]] <- factor(grille_f[[v]], levels = levels(df_freq[[v]]))
}

grille$freq_pred  <- predict(mod_freq_opt, newdata = grille_f,  type = "response")
grille$cout_pred  <- predict(mod_cout_opt, newdata = grille,    type = "response")
grille$prime_pure <- round(grille$freq_pred * grille$cout_pred, 2)

# Affichage
grille_aff <- grille[, c("CSP3","CLA3","CAR3","freq_pred","cout_pred","prime_pure")] |>
```

```
  mutate(freq_pred = round(freq_pred*100, 3),
         cout_pred = round(cout_pred, 2)) |>
  arrange(desc(prime_pure))

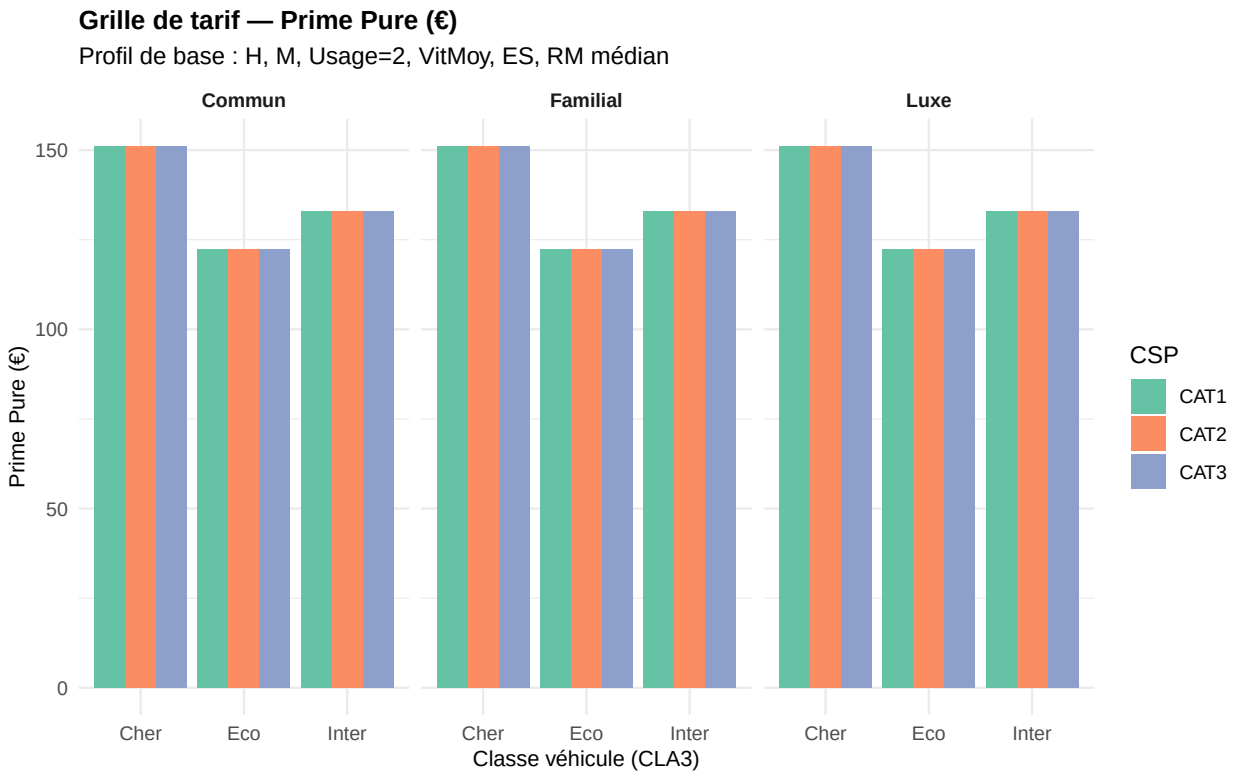
grille_aff |>
  kbl_out(caption = "Grille de tarif - Prime pure par profil (€)",
          col.names = c("CSP","Classe véhicule","Type véhicule",
                        "Fréq. (%)","Coût moy. (€)","Prime pure (€)"))
```

Table 11: Grille de tarif — Prime pure par profil (€)

CSP	Classe véhicule	Type véhicule	Fréq. (%)	Coût moy. (€)	Prime pure (€)
CAT1	Cher	Luxe	10.332	1463.47	151.20
CAT2	Cher	Luxe	10.332	1463.47	151.20
CAT3	Cher	Luxe	10.332	1463.47	151.20
CAT1	Cher	Commun	10.332	1463.47	151.20
CAT2	Cher	Commun	10.332	1463.47	151.20
CAT3	Cher	Commun	10.332	1463.47	151.20
CAT1	Cher	Familial	10.332	1463.47	151.20
CAT2	Cher	Familial	10.332	1463.47	151.20
CAT3	Cher	Familial	10.332	1463.47	151.20
CAT1	Inter	Luxe	9.094	1463.47	133.08
CAT2	Inter	Luxe	9.094	1463.47	133.08
CAT3	Inter	Luxe	9.094	1463.47	133.08
CAT1	Inter	Commun	9.094	1463.47	133.08
CAT2	Inter	Commun	9.094	1463.47	133.08
CAT3	Inter	Commun	9.094	1463.47	133.08
CAT1	Inter	Familial	9.094	1463.47	133.08
CAT2	Inter	Familial	9.094	1463.47	133.08
CAT3	Inter	Familial	9.094	1463.47	133.08
CAT1	Eco	Luxe	8.362	1463.47	122.38
CAT2	Eco	Luxe	8.362	1463.47	122.38
CAT3	Eco	Luxe	8.362	1463.47	122.38
CAT1	Eco	Commun	8.362	1463.47	122.38
CAT2	Eco	Commun	8.362	1463.47	122.38
CAT3	Eco	Commun	8.362	1463.47	122.38
CAT1	Eco	Familial	8.362	1463.47	122.38
CAT2	Eco	Familial	8.362	1463.47	122.38
CAT3	Eco	Familial	8.362	1463.47	122.38

```
ggplot(grille_aff, aes(x = CLA3, y = prime_pure, fill = CSP3)) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~ CAR3, ncol = 3) +
  labs(
    title    = "Grille de tarif - Prime Pure (€)",
    subtitle = "Profil de base : H, M, Usage=2, VitMoy, ES, RM médian",
    x = "Classe véhicule (CLA3)", y = "Prime Pure (€)", fill = "CSP"
  ) +
  scale_fill_brewer(palette = "Set2") +
```

```
  theme(strip.text = element_text(face = "bold"))
```



Lecture : La prime pure = P(sinistre) × coût moyen conditionnel. Elle est ensuite majorée d'un **coefficient de chargement** (frais de gestion, profit, sécurité) pour obtenir la **prime commerciale**.

---

5 Synthèse

Étape	Méthode	Résultat
Séparation	Seuil Q99	1384 attritionnels / 14 graves
Adéquation coût	Gamma (meilleur AIC)	Confirmé par KS
GLM Fréquence	Binomial/Logit + stepAIC	Variables discriminantes identifiées
GLM Coût	Gamma/Log + stepAIC	Multiplicateurs estimés
Prime pure	Fréquence × Coût	Grille tarifaire produite

Fichier de compte rendu — à rendre avec la copie.